

Triplet Loss-Based Concrete Crack Verification for Structural Health Monitoring and Digital Twin Applications

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Abstract—This paper proposes a novel deep-learning approach for verifying concrete cracks, leveraging a Siamese network with triplet loss. Inspired by FaceNet, the model is trained on an augmented "Crack dataset" containing rotations and shearing to enhance its ability to distinguish subtle crack variations. This method facilitates the evolution-centric monitoring of cracks, enabling early detection of potential failures and contributing to the development of digital twins for structural health monitoring. The model achieves high accuracy (97.36%) and precision (95.77%) in verifying crack patterns, demonstrating its potential for improving infrastructure maintenance and safety.

Index Terms—cracks verification; measurement; digital twins. Resnet

I. INTRODUCTION

The integrity of concrete structures is paramount to ensuring their longevity and safety. Traditional methods of crack detection are often manual, time-consuming, and subject to human error. With the advent of digitalization in the field of civil engineering, there is a growing need for automated and reliable methods of structural health monitoring [1]. This work introduces a deep learning-based model for the verification of cracks in concrete, leveraging the concept of triplet loss, a powerful tool in the realm of metric learning.

By continuously monitoring and analysing patterns, the model can help in the early detection of potential failures, thereby reducing the risk of catastrophic events and extending the lifespan of infrastructure. Furthermore, the model's capabilities align with the concept of digital twins—a digital replica of a physical entity that can be used for simulation, analysis, and control. By integrating the model into a digital twin framework, it would be able to create a dynamic and up-to-date representation of the structure's condition, enabling real-time decision-making and predictive maintenance strategies [2].

Incorporating the proposed approach and focusing on the concept of monitoring the evolution of cracks over time in civil structures, this paper proposes for the first time this concept with a novel methodology for crack verification and also with the perspective of identification of cracks. This method

This work is partially supported by the NATO Science for Peace and Security Programme Multy-Year Project G5924 - "Inspection and security by Robots interacting with Infrastructure digital twinS (IRIS)".

aims to precisely track and compare the characteristics of specific cracks over time, utilizing advanced image embedding techniques. By embedding images of the observed structure at different times, the approach leverages deep learning algorithms, specifically a Siamese network with ResNet [3], to generate high-dimensional embeddings that capture the unique features of each crack. This enables the identification of the exact same crack across multiple observations, facilitating a detailed analysis of its development. Such a capability is critical for assessing the structural health of civil infrastructures, allowing for timely interventions based on the progression of damage rather than just its presence. This evolution-centric perspective on crack analysis is pioneering, offering a more dynamic and informed approach to structural health monitoring. The proposed model is inspired by the FaceNet architecture [4], which has revolutionized face recognition by learning to encode faces into a compact embedding. Similarly, the proposed model encodes images of concrete surfaces into a feature-rich embedding space, where the distance between points corresponds to the similarity of the crack patterns they represent. The training dataset is derived from a "Crack dataset," [5] which has been enriched with data augmentation techniques such as shearing and rotations to generate triplets of images: an anchor (a reference image), a positive (an image with a similar crack pattern to the anchor), and a negative (an image with a different crack pattern). This approach is novel in the context of concrete crack verification and is particularly suited for the analysis of evolutionary patterns of cracks, which is critical for predictive maintenance and the assessment of structural health.

The performance of the proposed model will be evaluated using the validation loss, to assess generalization ability, and accuracy measures metrics like precision and recall to determine its effectiveness in correctly verifying crack patterns. This evaluation is crucial for ensuring the model's reliability and practical applicability in automated structural health monitoring.

The research comprises five sections: Related Works (II) surveys relevant literature, Methods and Procedures (III) explains the research approach, Implementation (IV) details the

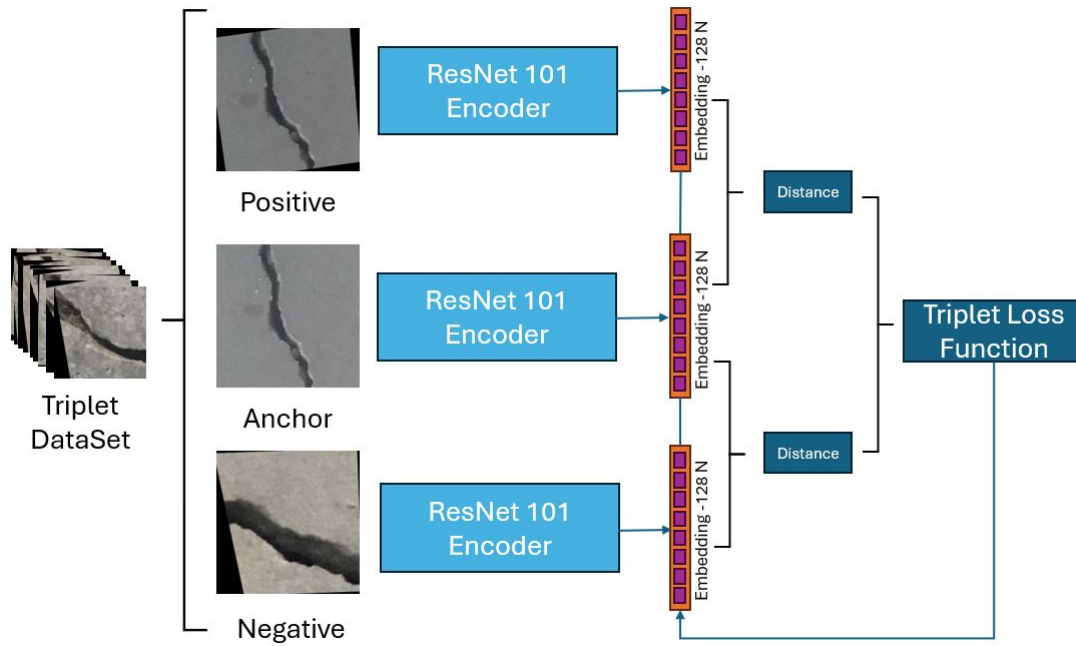


Fig. 1. Diagram of a Siamese network for crack verification, depicting the triplet dataset input into three ResNet-101 encoders to produce embeddings, which are then compared to compute triplet loss.

execution, Evaluation, and Discussion (V) analyzes outcomes, and Conclusions (VI) summarizes findings and suggests future research directions.

II. RELATED WORKS

The development and application of deep learning models for structural health monitoring (SHM) and the integration of these technologies into digital twin frameworks have been the focus of extensive research in recent years. This section reviews the literature relevant to the proposed method, which introduces a novel approach to verifying cracks in concrete structures using a deep learning model. The concept of using triplet loss [4] for enhancing the performance of deep learning models in recognizing patterns has been explored in various contexts. A notable application is the CrackEmbed model, which employs triplet loss for crack segmentation on the S3DIS dataset, demonstrating the utility of neural network embeddings in identifying crack patterns on 3D asphalt surfaces [6]. The integration of digital twin technology in civil engineering has been identified as a transformative approach toward more efficient and accurate infrastructure monitoring and maintenance. Recent literature highlights the development of digital twins ranging from static 3D models to fully integrated systems for real-time monitoring and control of civil infrastructure [7], [8]. The proposed work utilizes Siamese networks and ResNet 101 as the backbone for the proposed deep learning model [3]. Siamese networks, known for their effectiveness in computing similarity functions, are particularly suited for applications requiring the comparison of images or

patterns [9]. The use of ResNet 101, with its deep residual learning framework, further enhances the model's ability to learn from complex data structures and improve accuracy in crack detection and classification [10]. This combination of Siamese networks and ResNet 101 represents a powerful approach to addressing the challenges of automated crack detection in concrete structures. The distinction of this paper from other referenced works lies in its innovative methodology for crack verification in concrete structures through deep learning. Unlike conventional approaches that may rely on static data or simple pattern recognition, this study introduces an advanced technique by incorporating images of the observed structure captured at different times into the analysis. Utilizing a combination of Siamese networks and ResNet 101, the model generates high-dimensional embeddings. These embeddings are crafted to encapsulate the distinct features of each crack, leveraging the strength of Siamese networks in computing similarity functions and the deep residual learning capabilities of ResNet 101.

III. METHODS AND PROCEDURES

The methodology employs a deep learning-based approach to generate and analyze high-dimensional embeddings of images capturing structural cracks. This process facilitates the verification of specific cracks across different points, allowing for an accurate assessment of their evolution in the future. The core of our technique utilizes a Siamese network architecture, with ResNet101 serving as the backbone for feature extraction. This choice is motivated by ResNet101's proven capability to

capture intricate details within images, which is essential for differentiating between minor discrepancies in crack appearances [2]. ResNet101 is a deep neural network architecture with 101 layers, utilizing residual connections to facilitate training deep models by addressing the vanishing gradient problem, enhancing performance in image classification and feature extraction tasks (Fig. 1). The process begins with the collection of crack images of the targeted civil structure at various intervals. These images are then pre-processed to normalize their size and enhance contrast, ensuring that the input data is consistent and highlighting the features relevant to crack detection. For training the Siamese network, a triplet loss function has been employed [11]. This involves selecting triplets of images for each training iteration: an anchor (a reference image of a crack), a positive (another image of the same crack, possibly at a different time), and a negative (an image of a different crack). The network is trained to minimize the distance between the anchor and the positive images in the embedding space while maximizing the distance between the anchor and the negative images. This approach ensures that the network learns to identify the unique characteristics of each crack, making it possible to track the same crack over time through changes in its appearance. Once trained, the network can generate embeddings for new images of cracks. By comparing these embeddings, we can determine whether two images depict the same crack, even if there have been changes due to the crack's progression or variations in imaging conditions (Fig. 2). This capability allows for the detection of subtle changes in crack dimensions in future work that may indicate significant developments in the structure's condition.

The triplet loss function is used in machine learning to measure the relative similarity between inputs. The goal is to make the distance between the anchor and the positive smaller than the distance between the anchor and the negative by a

margin. The triplet loss equation is:

$$L = \max(0, d(a, p) - d(a, n) + \text{margin})$$

where $d(a, p)$ is the distance between the anchor (a) and the positive (p), $d(a, n)$ is the distance between the anchor and the negative (n), and margin is a threshold parameter to ensure the positive is closer to the anchor than the negative is by some margin. This approach is widely used in tasks like face recognition and similarity learning, encouraging a model to learn useful embeddings.

IV. IMPLEMENTATION

In the implementation of our methodology for crack verification involves several key stages, including image preprocessing, dataset creation, network configuration, and training processes, each characterized by specific hyperparameters and settings crucial for the system's effectiveness.

A. Image Preprocessing and Dataset

The dataset used in this study is the "Crack Dataset" available on Kaggle. This dataset is specifically designed for Crack Detection, providing a valuable resource for developing and testing algorithms in this field. The "Crack Dataset" contains a total of 20,000 images, each of size 227x227 pixels [12], [13]. Each entry in the dataset represents an instance of a crack, captured under various conditions. This diversity in data helps in creating robust models that can generalize well to real-world scenarios. For the study, the dataset has been transformed into a triplet set (anchor, positive, and negative) to fit the requirements of our model - a Siamese network with the triplet loss (Fig. 3). To provide the positive images, we used anchors with rotation and shearing. The dataset was divided as follows:

- 12,000 images for training (60%)
- 3,000 images for validation (15%)
- 5,000 images for testing (25%)

In this study, we utilized this dataset to train our model, validate its performance, and finally test its ability to generalize to unseen data. The use of a large testing set is crucial in the context of verification, as it provides a comprehensive assessment of the model's generalization capabilities to unseen data. It also aids in mitigating overfitting by ensuring that the model is evaluated against a significant amount of data that was not present during the training phase. The results, as discussed in the following sections, demonstrate the effectiveness of our approach and the potential of the "Crack Dataset" as a resource for crack detection research.

B. Network Configuration

The pre-trained ResNet101 architecture, provided by Keras [14] and trained on the ImageNet dataset, serves as the basis for our embedding model. This choice leverages the deep residual learning framework to facilitate the extraction of rich feature representations from images of civil structures. The network's output is then passed through additional dense layers

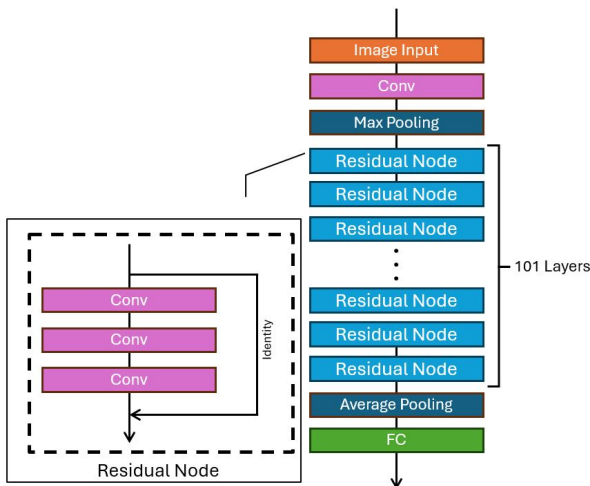


Fig. 2. Architecture of ResNet-101, highlighting the input, convolutional layers, residual nodes with skip connections (Identity), and the final Fully Connected (FC) layer.

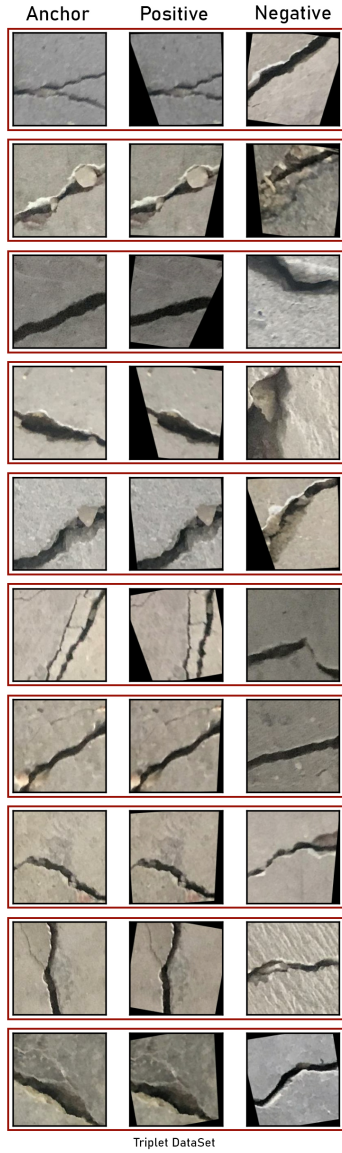


Fig. 3. Samples of triplet dataset for cracks.

and batch normalization layers to refine the feature embeddings. Specifically, we implement a dense layer with 256 units followed by ReLU activation [15] and batch normalization, and another dense layer with 128 units, also followed by ReLU activation and batch normalization, culminating in an output dense layer that produces 128-dimensional embeddings. The training process was done by utilizing V100 Nvidia GPU [16] and The validation loss model has reached $1.3e-05$ (Fig. 4).

C. Hyperparameters Summary

Here, outlines key configuration parameters for training the neural network, including image size, batch size, epochs, and optimizer details. Parameters such as learning rate, epsilon, and the margin for Triplet Loss are specified to optimize model performance. Hyperparameters as follows:

- Target Image Size: 227×227 pixels

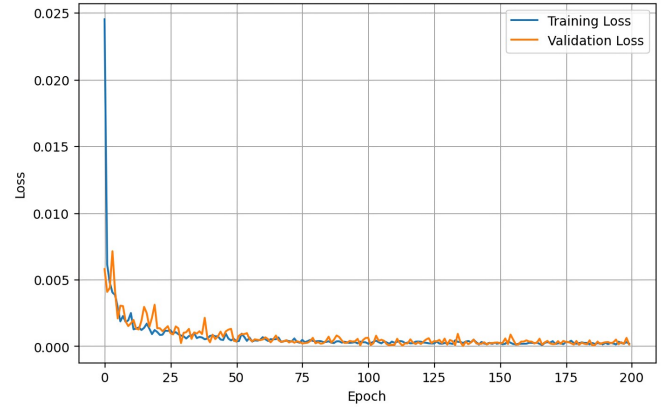


Fig. 4. Model Loss in during Training.

- Batch Size: 32
- Epochs: 200
- Optimizer: Adam [17]
- Learning Rate: $1e-4$
- Epsilon: $1e-1$
- Margin (for Triplet Loss): 1
- Dense Layer Configuration: 256 units (ReLU + Batch-Norm) followed by 128 units (ReLU + BatchNorm)

V. EVALUATION

The implementation of the Siamese Network with a ResNet-101 backbone for the task of image verification yielded significant insights into the model's ability to discern between similar and dissimilar images. The analysis of the distances between anchor-positive and anchor-negative pairs, derived from a test dataset comprising 5,000 image triplets, forms the crux of our evaluation.

A. Model Evaluation Metrics

To evaluate the performance of the proposed Siamese network model, a set of metrics has been utilized, each providing insights into different aspects of model accuracy and robustness [18].

- **Precision (P):** Precision measures the accuracy of the positive predictions made by the model. It is defined as the ratio of True Positives (TP) to all positive predictions (the sum of True Positives and False Positives).

$$P = \frac{TP}{TP + FP}$$

Where, FP is the number of False Positives. High precision indicates a low false positive rate.

- **Recall (R):** Recall, also known as sensitivity or true positive rate, measures the model's ability to identify all relevant instances. It is defined as the ratio of True Positives to the total number of actual positives (the sum of True Positives and False Negatives).

$$R = \frac{TP}{TP + FN}$$

Where, FN is the number of False Negatives. High recall indicates that the model is able to identify most of the positive instances.

- **F1 Score:** The F1 Score is the harmonic mean of precision and recall, providing a single metric that balances both concerns. It is especially useful when the class distribution is imbalanced.

$$F1 = \frac{2 \times P \times R}{P + R}$$

A high F1 score indicates that the model has both high precision and high recall.

- **Manhattan Distance (D):** Manhattan Distance, also known as L1 distance, measures the distance between two points in a grid-based path (as opposed to Euclidean distance). It is used to calculate the distance between embeddings in the Siamese network, reflecting the model's discriminative power.

$$D = \sum_{i=1}^n |x_i - y_i|$$

Where x_i and y_i are the components of the two points in the embedding space. Lower Manhattan distances indicate that the embeddings of similar instances are close to each other.

- **Standard Deviation (σ):** Standard deviation measures the dispersion or variability in a set of values. In the context of model evaluation, it can indicate the variability in the distance between embeddings, which reflects how consistently the model is able to map similar instances close together in the embedding space.

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$$

Where N is the number of observations, x_i is each individual observation, and $\bar{x}$ is the mean of the observations. A lower standard deviation indicates that the distances are more consistent.

B. Evaluation Model

Statistical analysis revealed that the mean distance for anchor-positive pairs was significantly lower than that for anchor-negative pairs (Fig. 5), illustrating the model's efficacy in embedding similar images closer in the feature space. Specifically, the mean distance for positive pairs was observed at 5.13, with a standard deviation of 2.01, indicating a tight clustering of similar images. Conversely, the mean distance for negative pairs stood at 19.89, with a standard deviation of 5.75, reflecting a broader dispersion that is expected given the dissimilarity among the images. Further examination through the determination of an optimal threshold for classification underscored the model's precision in distinguishing between the two categories. An optimal threshold of 10.56 was identified, balancing sensitivity and specificity, and enabling the model to achieve an accuracy of 97.36% (Fig. 6, Fig. 7). The precision

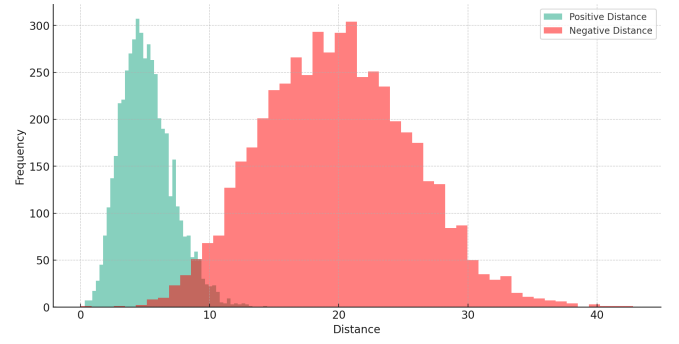


Fig. 5. The frequency distribution of distances for both positive and negative pairs.

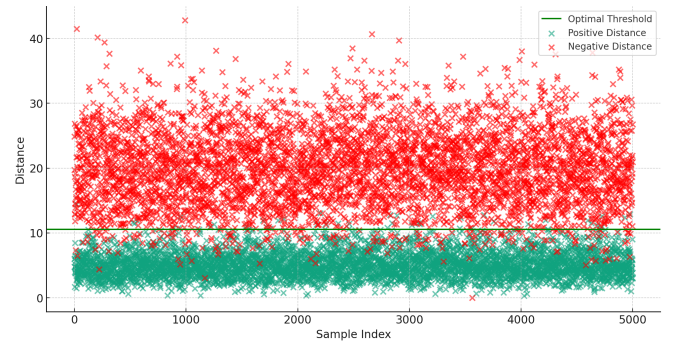


Fig. 6. Individual positive and negative distances, highlighting the spread and overlap of distances. The optimal threshold line shows the distance value that best separates positive from negative pairs.

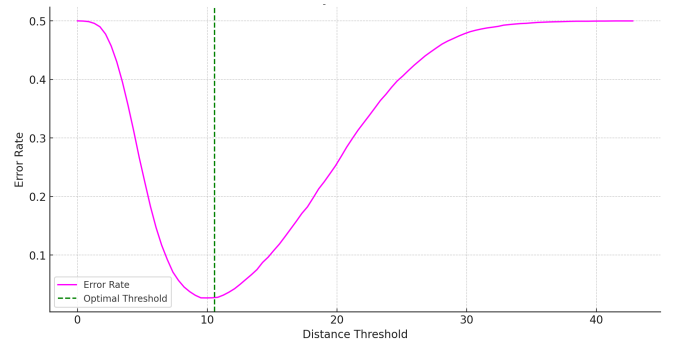


Fig. 7. This shows how the error rate of classification changes as the distance threshold is adjusted. The optimal threshold is marked, indicating the point at which the error rate is minimized.

and recall metrics, standing at 95.77% and 99.1% respectively, along with an F1 score of 97.41%, attest to the model's robustness and reliability in performing image verification tasks. The Area Under the Curve (AUC) for the Receiver Operating Characteristic (ROC) curve was calculated at 99.61%, highlighting the model's excellent discriminatory ability across various threshold settings (Fig. 8). The distribution plots for positive and negative distances further elucidated the model's discriminative capacity, showcasing a clear separation between

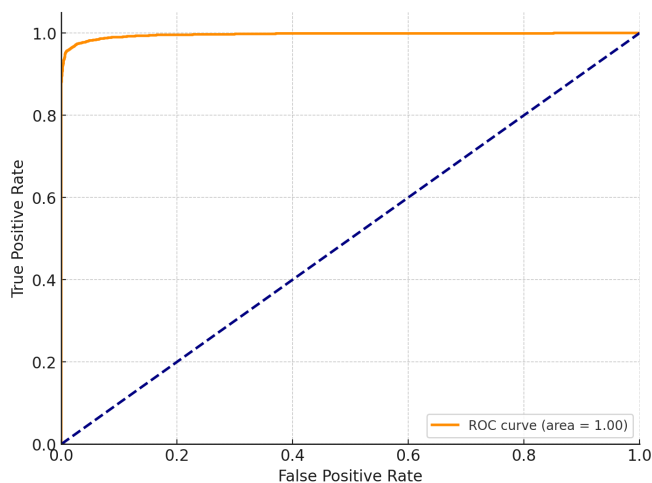


Fig. 8. The Receiver Operating Characteristic (ROC) curve above illustrates the performance of the Siamese network model across different threshold settings for the image verification task.

the two classes.

VI. DISCUSSION AND CONCLUSIONS

A. Discussion and Future Works

The results demonstrate the Siamese Network's potent capability to learn nuanced distinctions between images, leveraging the depth and sophistication of the ResNet-101 architecture. The high degree of accuracy and precision achieved suggests that deep learning, particularly Siamese Networks, holds substantial promise for the field of image verification and similar applications. One notable observation is the model's high recall rate, which implies that it is highly effective at identifying true positive pairs. This characteristic is especially valuable in scenarios where failing to recognize similar images carries significant consequences. The findings from this study underscore the importance of selecting an appropriate distance metric and threshold in optimizing the performance of Siamese Networks for specific tasks. The robustness of the chosen methodology, including the use of ResNet-101 as a feature extractor and the application of triplet loss for training, is validated by the high levels of accuracy and the discriminative precision observed in the results. A main limitation identified is the lack of use of real positive images in the triplet data, instead relying on modifications of the anchor image through shearing and rotation to create positive data. For further work, the incorporation of real positive crack images is recommended to potentially enhance model performance and generalizability.

B. Conclusions

This study demonstrates the effectiveness of a Siamese network with triplet loss for verifying concrete crack patterns. The model achieves high accuracy and precision, highlighting

its potential for applications in structural health monitoring. Future work could explore the integration of additional contextual information, alternative distance metrics, and different network architectures to further enhance the model's performance and generalizability. This research contributes to the development of deep learning-based approaches for infrastructure assessment, paving the way for more efficient and reliable maintenance practices.

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